MILITARY SPECIFICATION

VALVE, FLUID INJECTION, SOLENOID OPERATED

This specification is approved for use within the Army Missile Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification defines the requirements for one type solenoid operated on-off fluid injection valve.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

L-P-378	Plastic Sheet and Strip, Thin Gauge, Polyolefin
QQ-P-35	Passivation Treatments for Corrosion Resisting Steel

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Missile Command, ATTN:AMSMI-RD-SE-TD-ST, Redstone Arsenal, AL, 35898-5270 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) at the end of this document or by letter.

FSC 4810

AMSC N/A

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QQ-S-763	Steel Bars, Wires, Shapes and Forgings, Corrosion-Resisting
PPP-B-636	Boxes, Shipping, Fiberboard
PPP-C-1797	Cushioning Material, Resilient, Low Density, Unicellular, Polypropylene Foam
PPP-T-76	Tape, Packaging, Paper (For Carton Sealing)

MILITARY

MIL-P-116	Preservation, Methods of
MIL-B-117	Bags, Sleeves, and Tubing-Interior Packaging
MIL-T-152	Treatment, Moisture and Fungus-Resistant, of Communications, Electronic and Associated Electrical Equipment
MIL-S-7742	Screw Threads, Standard, Optimum Selected Series, General Specification for
MIL-I-23053	Insulation Sleeving, Electrical, Heat Shrinkable, General Specification for
MIL-P-25604	Propellant, uns-Dimethylhydrazine
MIL-S-45743	Soldering, Manual Type, High Reliability, Electrical and Electronic Equipment

STANDARDS

MILITARY

DOD-STD-100	Engineering Drawing Practices
MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U.S. Military Property
MIL-STD-143	Standards and Specifications, Order of Precedence for the Selection of

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MIL-STD-202	Test Methods for Electronic and Electrical Component Parts
MIL-STD-461	Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference
MIL-STD-462	Electromagnetic Interference Characteristics, Measurement of
MIL-STD-810	Environmental Test Methods and Engineering Guidelines
MIL-STD-889	Dissimilar Metals
MIL-STD-1523	Age Controls of Age Sensitive Elastomeric Materiel (For Aerospace Applications)
MIL-STD-45662	Calibration Systems Requirements
MS3116	Connectors, Plug, Electric, Series 1, Solder Type, Straight, Bayonet Coupling
MS20995	Wire, Safety or Lock
MS33540	Safety Wiring and Cotter Pinning, General Practices for
MS33656	Fitting End, Standard Dimensions for Flared Tube Connection and Gasket Seal

(Copies of specifications and standards required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets, or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order (see 4.4 and 6.2), a sample shall be subjected to first article inspection.

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3.2 Material. Materials, processes and parts shall be as specified herein. Materials not definitely specified shall be selected in accordance with the order of precedence specified in MIL-STD-143. Preference shall be given to selection of materials and processes such that a storage life of five years can be expected. The use of critical or strategic materials shall be kept to a minimum consistent with the design requirements (see 4.6.1).

3.2.1 Dissimilar metals. The use of dissimilar metals that tend to corrode when in contact, shall be avoided. Where the use of such metals cannot be avoided, they shall be suitably insulated from each other as defined in MIL-STD-889.

3.2.2 Corrosion resistant metals. All metals used shall be corrosion resistant or suitably protected to resist corrosion. Corrosion resistant steels shall be protected in accordance with QQ-P-35.

3.2.3 Use of magnesium. Magnesium shall not be used without prior approval of the procuring activity.

3.2.4 Threaded parts. All threaded parts shall conform to MIL-S-7742 and shall be positively locked or safetied in the installed position. Safety wire, where used, shall be applied in accordance with MS33540, shall conform to MS20995, and shall be covered with insulation sleeving conforming to MIL-I-23053/5, class I.

3.2.5 Soldering. Soldering of electrical connections shall be performed in accordance with MIL-S-45743.

3.2.6 Fungus proof materials. Materials which do not support the growth of fungi are preferable. If nutrient materials are used, they shall be treated in accordance with MIL-T-152.

3.2.7 Elastomeric parts. When elastomeric parts are used, they shall be compatible with operating fluids specified and all test and manufacturing fluids used. Marking shall be in accordance with MIL-STD-1523. Butyl elastomeric seals are not acceptable.

3.2.8 Lead wires. The coil lead wires shall be size 24 AWG.

3.2.9 High temperature exposure metals. All valve parts which extend past the plane of the mounting surface of the valve and into the injection port of an engine shall be of corrosion resistant steel conforming to QQ-S-763, type 347, condition A or a steel with equivalent high temperature properties of melting point, specific heat, thermal conductivity and density.

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3.3 Design and construction. The design and construction of the valve shall be in accordance with figure 1 (see 4.6.1).

3.3.1 Weight. The dry weight of the valve shall not exceed 1.1 pounds.

3.3.2 Alignment. All plungers, poppets, balls and pistons shall be accurately guided to prevent misalignment or chattering on their seats.

3.3.3 Valve operation. The valve shall provide rapid on-off control of liquid propellant under pressure that is injected into an engine thrust chamber. Valve on-off operation shall be accomplished by energizing an electromagnet. Current application shall cause the valve to open and current removal shall cause the valve to close.

3.3.4 Operating fluid. The valve operating fluid shall be unsynmetrical dimethylhydrazine (UDMH) conforming to MIL-P-25604. All volumetric flow rates are based on MIL-P-25604 fuel usage. An approved test fluid may be used for performance and evaluation unless otherwise specified.

WARNING

UDMH is a hazardous material and should be handled in accordance with the manufacturer's safety data sheet.

3.3.5 Fluid contamination. The valve shall perform as specified when the operating fluid is contaminated to the levels shown in table I.

TABLE I. Level of contaminants.

Particle size, microns	Particles per 100 milliliters 1/
<10	Unlimited
10-25	6000
26-50	2000
51-100	800
101-300	50
301-500	5
>500	0

1/ Maximum contaminants = 15 milligrams per liter.

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3.4 Performance requirements.

3.4.1 Flow rate. The valve shall flow 5.5 pounds per second (lb/sec) minimum and 5.7 lb/sec maximum of UDMH with a 900 pound per square inch (psi) pressure drop across the valve (see 4.6.2).

3.4.2 Pressure ratings.

3.4.2.1 Normal operating pressure. The valve shall operate as specified at 900 to 1100 pounds per square inch gauge (psig), (see 4.6.3.1).

3.4.2.2 Surge pressure. The valve shall withstand pressure surges from 0 to 2300 psig peak of 10 milliseconds minimum and 100 milliseconds maximum with no degradation of performance (see 4.6.3.2).

3.4.2.3 Proof pressure. The valve in the de-energized condition shall withstand a proof pressure of 1650 psig for 3 minutes. There shall be no external leakage, rupture, permanent deformation or degradation of performance (see 4.6.3.3).

3.4.2.4 Burst pressure. The valve in the de-energized condition shall withstand a burst pressure of 4500 psig for 3 minutes. There shall be no external leakage (see 4.6.3.4).

3.4.3 Initial pressurization operating characteristics. The valve full opening and full closing times at the specified pressure levels shall be as shown in table II, (see 4.6.4).

TABLE II. Pressure vs. full opening and closing times.

Pressure across valve (psi)	Opening and closing times (milliseconds)
400	15 ± 6
800	13 ± 4
900	See table III

3.4.4 Maximum installation temperature. The valve shall perform as specified when installed on an engine which is ignited and fired for a maximum duration of 6.5 seconds. The heat added to the valve through the engine firing shall be in addition to the initial temperature conditions of the valve and UDMH of -40° Celsius (C) to 72° C [(-40 Fahrenheit (F) to 160°F], (see 4.6.5).

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3.4.5 Internal leakage. The internal leakage shall not exceed 1.0 gallons per minute (gpm) of UDMH with the valve de-energized and with 1000 psig applied to the inlet (see 4.6.6).

3.4.6 External leakage. The external leakage shall be zero at any applied pressure up to 4500 psig maximum (see 4.6.7).

3.4.7 Injection stream shape. The injection stream shape, injected into the atmosphere with 900 ± 10 psi across the valve, shall have a maximum total included angle of 13° at a location 4.000 ± 0.004 inches downstream from the surface of the valve mounting flange (see 4.6.8).

3.4.8 Injection stream momentum. The momentum of the injection stream shall be a minimum of 93 percent of the ideal momentum with a pressure drop of 900 ± 10 psi across the valve. Ideal momentum is defined as:

$$F_I = Q\left[\frac{2W\ \Delta P}{g}\right] \text{ where}$$

F_I = Stream ideal momentum force lbf
Q = Valve volumetric flow in^3/sec
W = Fluid specific weight lbf/in^3
ΔP = Valve pressure drop lbf/in^2
g = Acceleration of gravity in/sec^2

The momentum of the injection stream shall be measured on a circular force plate 1.2715 ± 0.0005 inches in diameter located 4.000 ± 0.004 inches from and perpendicular to the surface of the valve mounting flange directly in line with the centerline of the flow metering pin (see 4.6.9).

3.4.9 Injection stream perpendicularity. The centerline of the injection stream shall not deviate by more than 0.5° from a line drawn perpendicular to the surface of the valve mounting flange and passing through the centerline of the flow metering pin, when the center of the load cell force plate is not more than 0.035 inch from the true center of the flow metering pin (see 4.6.10).

3.4.10 Flow stability. The flow stability of the valve shall be such that after 1000 cycles of operation, the injectant flow rate with 900 ± 10 psi across the valve and a regulated four hertz (Hz) square wave for 230 seconds applied, shall be within three percent of the injectant flow rate measured after 100 cycles of operation (see 4.6.11).

3.4.11 Operating time. Valve operating time, with 900 ± 10 psi across the valve, shall be as specified in table III (see 4.6.12).

3.4.12 Operating life. The valve shall have a minimum operating life of 10,000 cycles without malfunction (see 4.6.13).

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TABLE III. Valve operating time (milliseconds).

Temperature	Full on			Full off		
	Electrical delay 1/	Output flow rise time 1/	Total	Electrical delay 1/	Output flow decay time 1/	Total
16 ± 33°C (60 ± 60°F)	7 ± 2.5	4 ± 2	12 ± 3	7.5 ± 25	4 ± 2	11 ± 3
At -40°C (-40°F) and 72°C (160°F)	7 +3.5 -2.5	4 ± 2	12 +4 -3	7.5 +3. -2.	4 ± 2	11 +4 -3

1/ Electrical delay, flow rise, and decay times apply only to UDMH usage at the system test level. The total ON and OFF times apply to both UDMH system level usage and UDMH or UDMH simulation fluid usage at the component level.

3.5 Electrical characteristics.

3.5.1 Input voltage. The input voltage to the solenoid shall be 25.0 volts direct current (Vdc) minimum to 33.0 Vdc maximum (see 4.6.14.1).

3.5.2 Coil resistance. The solenoid shall have the temperature versus resistance characteristics shown in table IV (see 4.6.14.2).

TABLE IV. Temperature vs. resistance characteristics.

Temperature	DC resistance ohm
20°C (68°F)	36 + 1.8
72°C (160°F)	45.4 maximum
-40°C (-40°F)	26.0 minimum

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3.5.3 Input current. The valve shall operate when energized with input currents from 0.551 ampere (A) minimum to 1.27 A maximum (see 4.6.14.3).

3.5.4 Inductance. The inductance of the solenoid shall be 0.85 to 0.88 henries (H) over the temperature range of -40°C to 72°C (-40°F to 160°F) at frequencies of 0 to 22 cps (see 4.6.14.4).

3.5.5 Insulation resistance. The insulation resistance between electrical circuits and case shall be not less than 50 megohm at an applied potential of 500 Vdc for 60 seconds over all operational temperatures (see 4.6.14.5).

3.6 Environmental. After being subjected to the nonoperating environmental conditions, the valve shall meet the requirements of this specification while being subjected to the operating environmental conditions or any combination thereof (see 4.6.15).

3.6.1 Nonoperating.

3.6.1.1 Vibration. The valve, empty of fluid, shall withstand sinusoidal vibration from 10 to 28 Hz at 0.25 inch double amplitude (in da) and 28 to 500 Hz at 10 gravity units (g) peak for six hours in each of the three mutually perpendicular axes as shown on figure 1 (see 4.6.15.1.1).

3.6.1.2 Shock. The valve shall withstand three half-sine shock pulses of 450 g applied for three milliseconds in each of three mutually perpendicular axes as shown on figure 1 (see 4.6.15.1.2).

3.6.1.3 Altitude. The valve shall withstand exposure to any altitude from sea level to 50,000 feet for 12 hours (see 4.6.15.1.3).

3.6.1.4 Humidity. The valve shall withstand a stabilized relative humidity of 30 percent at a temperature of 21.2° ± 1.4°C (70° ± 2.5°F), (see 4.6.15.1.4).

3.6.1.5 Fungus. The valve shall be nonnutritive to fungus growth such as encountered in tropical climates (see 4.6.15.1.5).

3.6.1.6 Storage temperature. The valve shall withstand storage at temperatures from -54°C (-65°F) to 72°C (160°F), (see 4.6.15.1.6).

3.6.2 Operating.

3.6.2.1 Vibration. The valve shall withstand random vibration of 6.7 g^2/Hz from 50 to 1500 Hz attenuated at 24 decibels per octave on both high and low frequency ends within the overall bandwidth of 20 to 2000 Hz with a minimum root mean square (rms) acceleration for this spectrum of 100 g for 6.5 seconds along each of the three mutually perpendicular axes as shown on figure 1 (see 4.6.15.2.1).

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3.6.2.2 Shock. The valve shall withstand half-sine shock pulses of 600 g peak for 1.0 to 1.5 milliseconds in both plus and minus directions of each of the three mutually perpendicular axes as shown on figure 1 (see 4.6.15.2.2).

3.6.2.3 Altitude. The valve shall withstand exposure to any altitude from sea level to 50,000 feet (see 4.6.15.2.3).

3.6.2.4 Explosive atmosphere. The valve shall not cause ignition when operated in an ambient atmosphere of an explosive gaseous mixture (see 4.6.15.2.4).

3.6.2.5 Boost acceleration. The valve shall withstand acceleration levels applied for 6.5 seconds in accordance with the following (see 4.6.15.2.5):

a. Forward at 40.0 g.

b. Aft at 3.0 g.

c. Lateral and up at $\pm$ 2.0 g.

d. Angular about spin axes at 13.5 radians/sec^2.

e. Angular about transverse axes at $\pm$ 2.1 radians/sec^2 at the transverse moment arm of 107 inches.

3.6.2.6 Spin. The valve shall withstand any spin from 0 to 5.5 revolutions per second (rps) at a moment arm of 6.5 inches (see 4.6.15.2.6).

3.6.2.7 Acoustical noise level. The valve shall withstand sound pressure levels of 168 db applied over a bandwidth of 37.5 to 9600 Hz applied in a reverberant sound field for 6.5 seconds (see 4.6.15.2.7).

3.6.2.8 Temperature. The valve shall withstand environmental and operating fluid temperatures from -40°C to 72°C (-40°F to 160°F), (see 4.6.15.2.8).

3.6.3 Electromagnetic interference. Electrical circuits shall not cause radio interference during operation beyond the limits as specified in MIL-STD-461 and MIL-STD-462 (see 4.6.16).

3.7 Marking. The valve shall be marked in accordance with MIL-STD-130 (see 4.6.1).

3.8 Workmanship. There shall be no cracks, burrs, dents, loose attaching parts of misalignment or any other evidence of poor workmanship that would make the valve unsuitable for the purpose intended (see 4.6.1).

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Test equipment and inspection facilities. The manufacturer shall insure that test and inspection facilities of sufficient accuracy, quality, and quantity are established and maintained to permit performance of required inspections.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4)

b. Quality conformance inspection (see 4.5)

4.3 Inspection conditions. Unless otherwise specified, all tests shall be conducted under the following conditions:

a. Ambient temperature; 26.7° ± 16.6°C (80° ± 30°F).

b. Test fluid temperature; 32.2° ± 11.1°C (90° ± 20°F).

c. Relative humidity; 90 percent or less.

d. Barometric pressure; 30 ± 2 inches of mercury (in Hg).

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4.3.1 Inspection equipment. The quality and performance of the equipment used in conducting the inspections specified shall be compatible with the requirements of inspections to be performed and shall be controlled by a test equipment calibration system in accordance with MIL-STD-45662.

4.4 First article inspection.

4.4.1 First article sample. When specified in the contractor purchase order (see 3.1 and 6.2), the first article sample shall consist of the number of samples specified by the procuring activity and shall be representative of the production product. The first article sample shall be subjected to all inspections and tests as specified under 4.6 and shall be conducted in an order acceptable to the procuring activity. Quantity production shall be withheld until the first article sample has been pronounced satisfactory by the procuring activity.

4.5 Quality conformance inspection. Quality conformance inspection for acceptance shall consist of the inspection and tests specified in table V in an order acceptable to the procuring activity.

TABLE V. Quality conformance inspection.

Characteristics	Requirement paragraph	Method paragraph	Frequency of inspection
Examination of product----	3.2, 3.3, 3.7, and 3.8	4.6.1	100 percent
Flow rate----------	3.4.1	4.6.2	100 percent
Normal operating pressure-	3.4.2. 1	4.6.3.1	100 percent
Surge pressure----------	3.4.2. 2	4.6.3.2	
Proof pressure----------	3.4.2. 3	4.6.3.3	
Maximum installation temperature----------	3.4.4	4.6.5	
Internal leakage----------	3.4.5	4.6.6	100 percent
External leakage----------	3.4.6	4.6.7	100 percent
Injection stream shape----	3.4.7	4.6.8	
Injection stream momentum-	3.4.8	4.6.9	
Injection stream perpendicularity---------	3.4.9	4.6.10	
Operating time----------	3.4.11	4.6.12	100 percent
Coil resistance----------	3.5.2	4.6.14.2	100 percent
Insulation resistance----	3.5.5	4.6.14.5	100 percent

4.5.1 Sampling inspection. Sampling inspection shall be in accordance with MIL-STD-105. An acceptable quality level (AQL) of 1.0 percent shall be used for major characteristics and an AQL of 4.0 percent shall be used for minor characteristics. One hundred percent inspection shall be used when required.

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4.5.2 Noncompliance. Failure of a valve to comply with the requirements specified shall be cause for rejection of individual units or lots, as applicable. Valves may be reworked to correct defects and resubmitted for inspection. However, before reinspection, the manufacturer shall notify the procuring activity and the cognizant inspection activity of such failure. Acceptance and shipment of the product shall be discontinued until corrective action acceptable to the procuring activity has been taken. In the event of failure after reinspection, information concerning the failure shall be furnished to the congnizant inspection activity and the procuring activity.

4.5.3 Inspection and packaging. Preservation, packing and marking shall be inspected for conformance to section 5.

4.6 Methods of inspection.

4.6.1 Examination of product. Materials and processes used in the manufacture of the valve shall be verified for conformance to this specification. The valve shall be visually and dimensionally inspected for conformance to figure 1 and weighed (see 3.2, 3.3, 3.7, and 3.8).

4.6.2 Flow rate. The valve shall be cycled four times from zero to full current at 0.25 Hz and the flow rate measured for conformance to 3.4.1.

4.6.3 Pressure ratings.

4.6.3.1 Normal operating pressure. The valve shall be pressurized and tested for conformance to 3.4.2.1.

4.6.3.2 Surge pressure. The valve shall be pressurized and tested for conformance to 3.4.2.2.

4.6.3.3 Proof pressure. The valve shall be pressurized to proof pressure and inspected for conformance to 3.4.2.3.

4.6.3.4 Burst pressure. The valve shall be pressurized to burst pressure and inspected for conformance to 3.4.2.4.

4.6.4 Initial pressurization operating characteristics. The valve shall be pressurized and the full opening and full closing times measured for conformance to 3.4.3.

4.6.5 Maximum installation temperature. The valve shall be installed on an engine and tested for conformance to 3.4.4. The quantity of valves to be tested and the quantity of engines required shall be as specified by the procuring activity (see 6.2).

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4.6.6 Internal leakage. The valve shall be pressurized and tested for conformance to 3.4.5.

4.6.7 External leakage. The valve shall be monitored for external leakage during all functional testing (see 3.4.6).

4.6.8 Injection stream shape. The valve shall be tested for conformance to 3.4.7.

4.6.9 Injection stream momentum. The valve shall be tested for conformance to 3.4.8.

4.6.10 Injection stream perpendicularity. The valve shall be tested for conformance to 3.4.9.

4.6.11 Flow stability. The valve flow stability shall be tested for conformance to 3.4.10.

4.6.12 Operating time. Six input current signals shall be applied and the valve tested for conformance to 3.4.11. The delay, flow rise and decay time shall be determined with the valve installed as part of an engine. The quantity of valves to be tested and the quantity of engines required shall be as specified by the procuring activity (see 6.2).

4.6.13 Operating life. The operating life test shall be a continuation of the flow stability test. The input signal shall be regulated to provide a 33 $\pm$ 2 Hz square wave signal to cycle the valve for conformance to 3.4.12.

4.6.14 Electrical characteristics.

4.6.14.1 Input voltage. The valve shall be tested for conformance to 3.5.1.

4.6.14.2 Coil resistance. The coil resistance across pins A and B of the electrical connector shall be measured in accordance with MIL-STD-202, method 303 for conformance to 3.5.2.

4.6.14.3 Input current. The valve shall be energized and tested for conformance to 3.5.3.

4.6.14.4 Inductance. The inductance of the valve coil shall be measured for conformance to 3.5.4.

4.6.14.5 Insulation resistance. The insulation resistance of the valve coil shall be measured in accordance with MIL-STD-202, method 302, test condition B for conformance to 3.5.5.

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4.6.15 Environmental. The valve shall meet the requirements of this specification after being subjected to the nonoperating environmental conditions and while being subjected to the operating environmental conditions or any natural combination thereof (see 3.6).

4.6.15.1 Nonoperating.

4.6.15.1.1 Vibration. The valve shall be tested in accordance with MIL-STD-810, method 514 for conformance to 3.6.1.1.

4.6.15.1.2 Shock. The valve shall be tested in accordance with MIL-STD-810, method 516 for conformance to 3.6.1.2.

4.6.15.1.3 Altitude. The valve shall be tested in accordance with MIL-STD-810, method 500, procedure I for conformance to 3.6.1.3.

4.6.15.1.4 Humidity. The valve shall be tested in accordance with MIL-STD-810, method 507, procedure II for conformance to 3.6.1.4.

4.6.15.1.5 Fungus. The valve shall be tested in accordance with MIL-STD-810, method 508 for conformance to 3.6.1.5.

4.6.15.1.6 Storage temperature.

a. High temperature. The valve shall be tested in accordance with MIL-STD-810, method 501, procedure I for conformance to 3.6.1.6.

b. Low temperature. The valve shall be tested in accordance with MIL-STD-810, method 502, procedure I for conformance to 3.6.1.6.

4.6.15.2 Operating.

4.6.15.2.1 Vibration. The valve shall be tested in accordance with MIL-STD-810, Method 514 for conformance to 3.6.2.1.

4.6.15.2.2 Shock. The valve shall be tested in accordance with MIL-STD-810, method 516, procedure IV for conformance to 3.6.2.2.

4.6.15.2.3 Altitude. The valve shall be tested in accordance with MIL-STD-810, method 500, procedure II for conformance to 3.6.2.3.

4.6.15.2.4 Explosive atmosphere. The valve shall be tested in accordance with MIL-STD-810, method 511, procedure I for conformance to 3.6.2.4.

4.6.15.2.5 Boost acceleration. The valve shall be tested in accordance with MIL-STD-810, method 513, procedure II for conformance to 3.6.2.5.

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4.6.15.2.6 Spin. The valve shall be tested for conformance to 3.6.2.6.

4.6.15.2.7 Acoustical noise level. The valve shall be tested in accordance with MIL-STD-810, method 515 for conformance to 3.6.2.7.

4.6.15.2.8 Temperature.

a. High temperature. The valve shall be tested in accordance with MIL-STD-810, method 501, procedure II for conformance to 3.6.2.8.

b. Low temperature. The valve shall be tested in accordance with MIL-STD-810, method 502, procedure II for conformance to 3.6.2.8.

4.6.16 Electromagnetic interference. The valve shall be tested for conformance to 3.6.3.

5. PACKAGING

5.1 Preservation. Preservation shall be level A.

5.1.1 Cleaning. Valves shall be cleaned so that the contamination added by the valve to the purge fluid, when 100 milliliters of purge fluid is passed through the unit under pressure, shall not exceed the limits of table I. The purge fluid shall be compatible with all materials used in the construction of the unit and the operating fluid.

5.1.2 Drying. Drying shall be in accordance with MIL-P-116.

5.1.3 Preservative application. Preservatives shall not be used.

5.1.4 Unit packs. Each valve shall be individually unit packed in contamination barrier material of polyethylene plastic film conforming to L-P-378 or polyethylene plastic bags conforming to MIL-B-117. The film or bags shall be closed by either heat sealing or with tape.

5.2 Packing. Packing shall be level A.

5.2.1 Level A. Valves preserved as specified in 5.1 shall be packed in containers conforming to PPP-B-636, style RSC, type CF, grade 175, class DOM and cushioned with polypropylene foam material conforming to PPP-C-1797. Container closure shall be in accordance with PPP-T-76 with barrier in accordance with MIL-B-117.

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5.3 Marking.

5.3.1 Unit packs. Valves which have been preserved as specified in 5.1 shall have a label attached which contains, as a minimum, the following:

> CAUTION. This item has been specially cleaned and sealed in a contamination barrier. Do not remove contamination barrier until ready for installation.

5.3.2 Exterior containers. In addition to any special or other identification marking required by the procuring activity (see 6.2), each container shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The valve is intended for use in a small liquid propellant rocket engine to regulate the UDMH fuel injection in the secondary thrust vector control.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number and date of this specification.

b. Whether first article is required (see 3.1 and 4.4) and number of samples required (see 4.4.1).

c. Quality conformance inspection if other than specified (see 4.5.1).

d. Quantity of valves and engines required (see 4.6.5 and 4.6.12).

e. Applicable level of preservation, packing and marking if other than specified (see section 5).

6.3. First article. When a first article inspection is required, the item should be a first article sample. The first article should consist of the number of units specified by the procuring activity. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, approval of first article test results and disposition of first article. Invitations for bids should provide that the Government reserves the right to waive the requirements for samples

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for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.4 Metrication. Metric equivalents in accordance with FED-STD-376 are acceptable for use in this specification.

6.5 Subject term (keyword) listing.

Valve, fluid injection, solenoid operated
Valve, solenoid operated

Custodian:
Army - MI

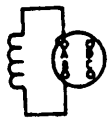
Preparing activity:
Army - MI

Review activities:
DLA - CS

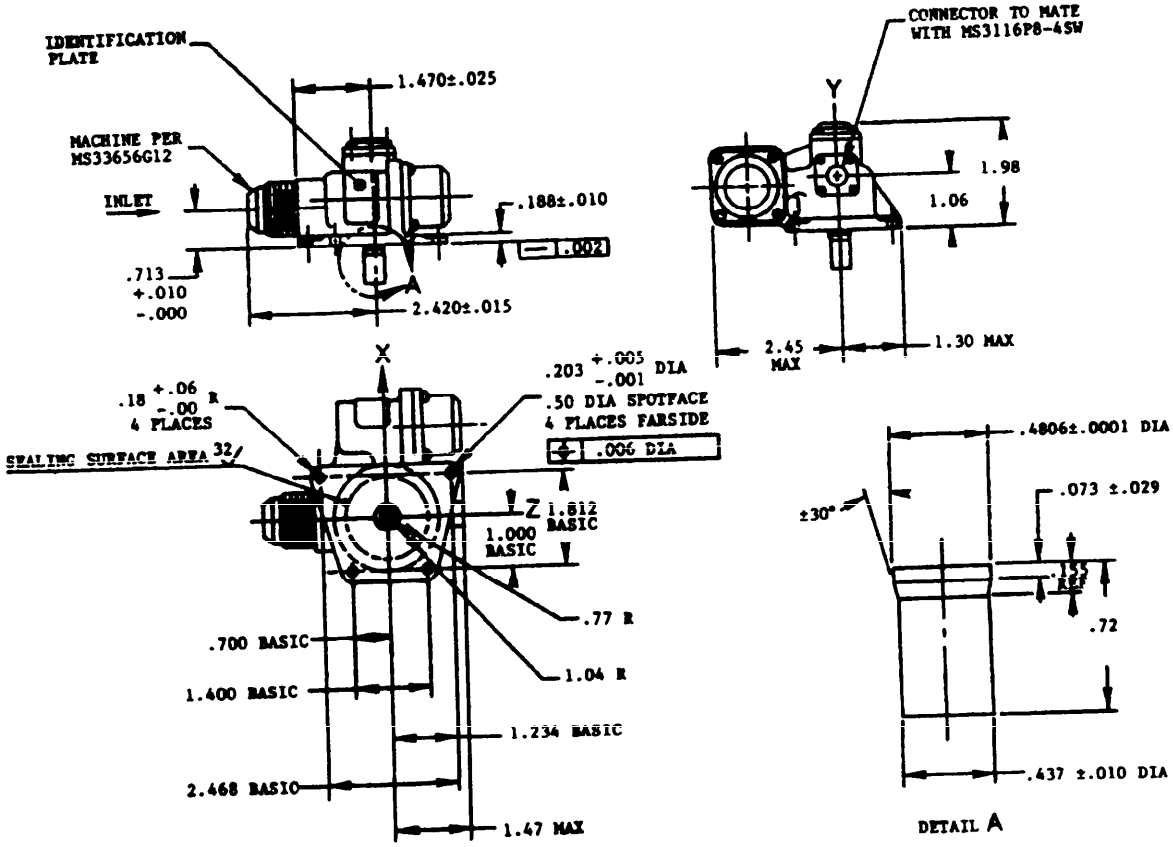
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VALVE AXIS	MISSILE AXIS
X	LONGITUDINAL
Y	RADIAL
Z	TRANSVERSE



WIRING DIAGRAM
(NORMALLY CLOSED WHEN DE-ENERGIZED)



NOTES:

1. All dimensions are in inches.
2. Metric equivalents shall be acceptable.

FIGURE 1. Valve, fluid injection, solenoid operated.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-V-47320	Valve, Fluid Injection, Solenoid Operated

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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